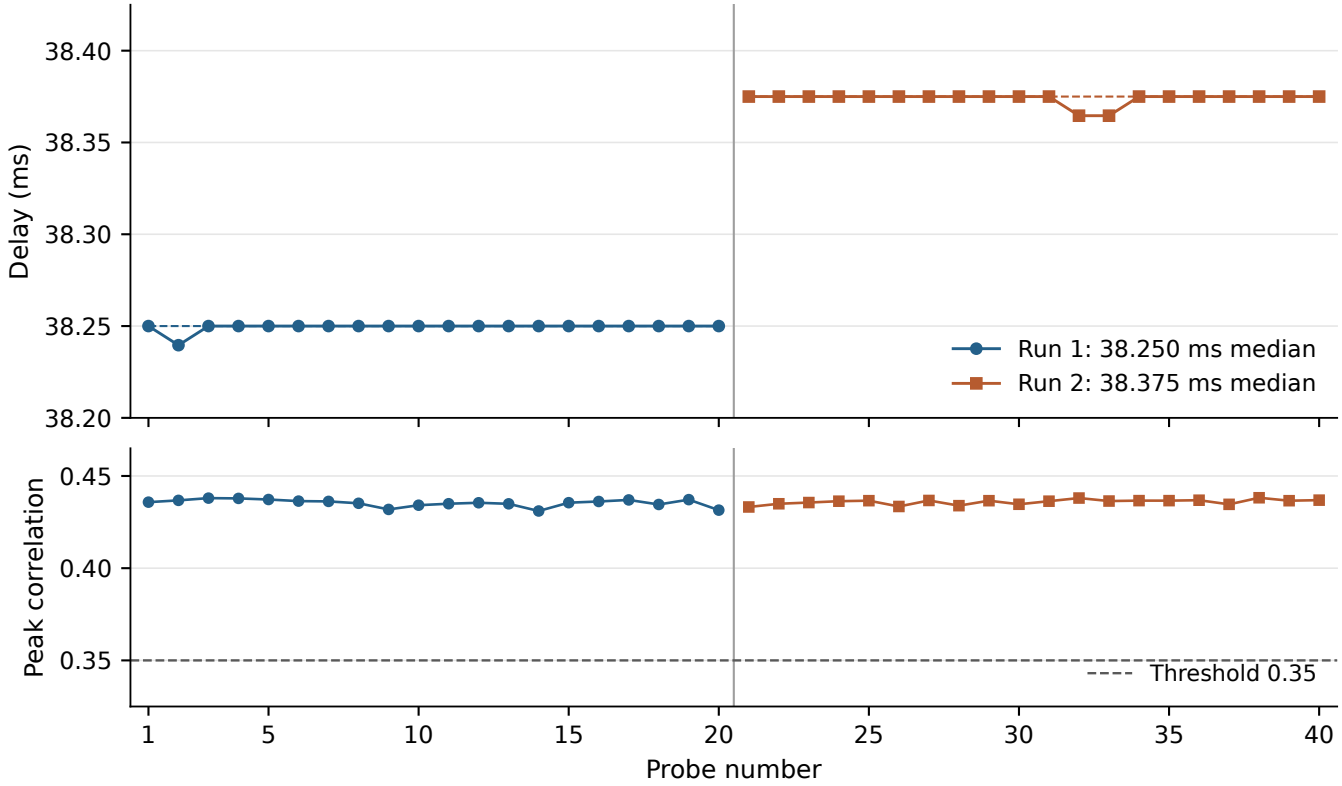
Playback-to-microphone delay

Audient iD14 · Edifier speaker · Earthworks M50

Pooled median	Observed spread	Accepted sweeps
38.307 ms	**0.135 ms**	**40 / 40**

Recorded at 96 kHz, with approximately 10 cm microphone separation (operator reported). The range was 38.240–38.375 ms across two stream starts, with 20 sweeps each.



Delay estimates above; waveform matching strength below. The dashed strength threshold is 0.35. Probes were separate 100 Hz–20 kHz timing sweeps.

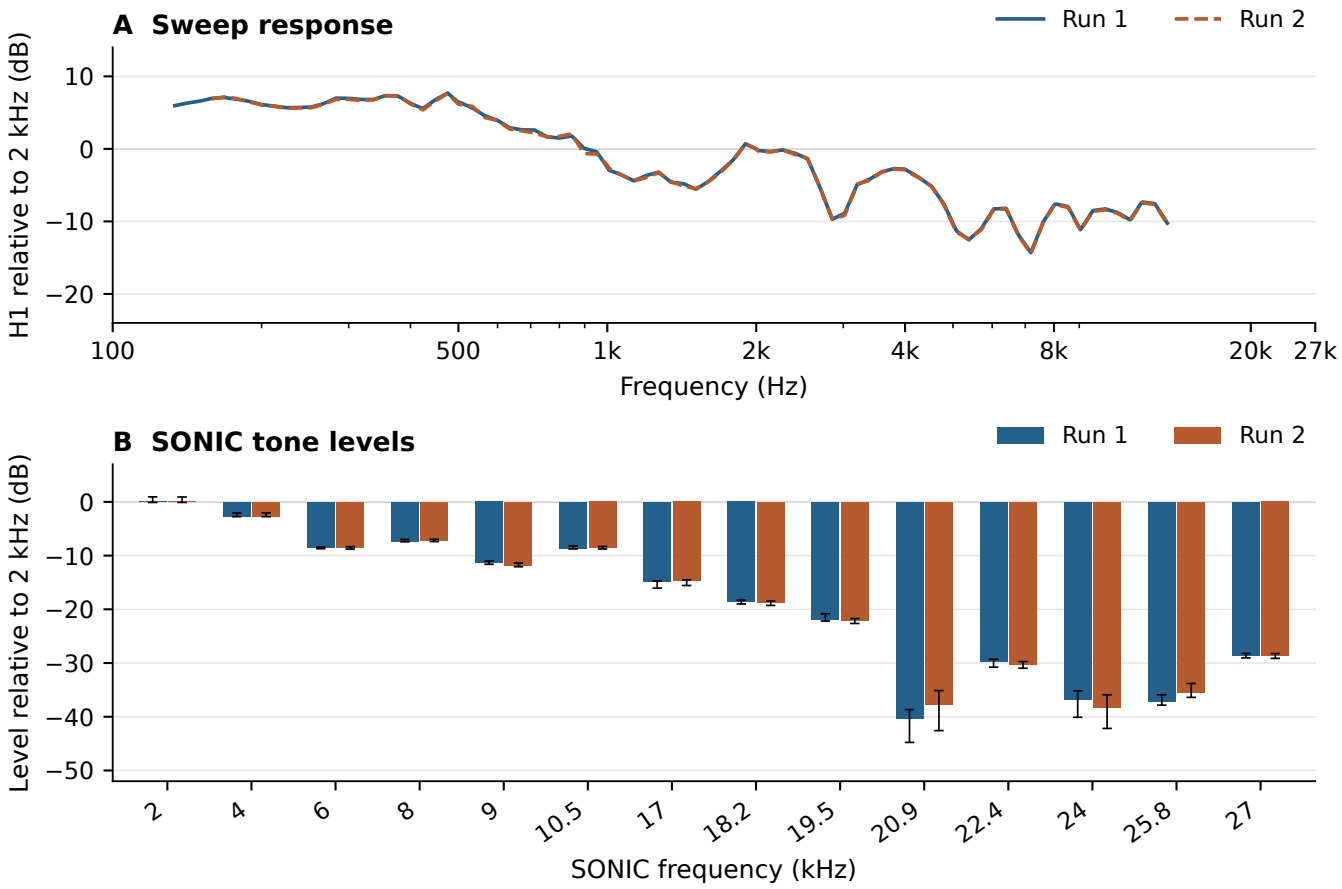
Interpretation. Timing was repeatable over these short runs; stream medians differed by 0.125 ms. This application roundtrip includes buffering and the acoustic path. The spread does not establish calibrated timing accuracy, isolated speaker delay or neural latency.

Recorded frequency levels

SONIC tone reconstruction · 14 frequencies · 2–27 kHz

Tone duration	Ramp duration	Recorded pips
10 ms	**2 ms**	**16,800**

Locally reconstructed Sheep E tones in two 84-second blocks.



Gaps mark unsupported sweep bands. Tone bars show medians; whiskers are 5th–95th percentiles of 600 pips per frequency per recording, not confidence intervals.

Finding. All 14 components appear in both recordings at unequal levels. Relative to 2 kHz, 17–19.5 kHz was 15–22 dB lower; 20.9–25.8 kHz was 30–40 dB lower. These uncorrected full-chain measurements leave SPL, speaker-only response and distortion unqualified.

Next. Fix the geometry, calibrate microphone/SPL, test separated tones, and validate timing with an independent electrical reference.